

GetPrep

**A PROJECT REPORT
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the Requirements for the Degree
of**

MASTER OF COMPUTER APPLICATION

**Under the Supervision of
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have carried out the project work having “**GetPrep**” (**Major Project, CA401P**)
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ABSTRACT

The GetPrep is an AI Mock Interview Web Application is a web-based, smart platform that assists job applicants in improving their interview performance using real-time simulations powered by AI. With the highly competitive nature of the job market today, there is a pressing need to be well prepared for interviews to secure a job. The conventional mock interview process can be expensive and time-consuming since it involves human mentors.

The site creates **dynamic, profession-specific interview questions** based on the user's role and level of experience. Users can answer in text or voice, and AI-based assessment offers immediate feedback on factors like clarity, confidence, technical correctness, and aptness. The system also features performance monitoring, analytics, and customized improvement recommendations, enabling users to hone their answers with repeated practice sessions.

The most important goals of this project are automated generation of interview questions, response **assessment based on AI**, and **adaptive learning processes** to make users more prepared. Through the offering of an **affordable, cost-efficient, and AI-driven** alternative to conventional mock interviews, this platform equips users with the skillset and confidence needed to succeed in actual job interviews, thus driving their chances of getting hired.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
GetPrep	Career Readiness & AI-based Candidate Knowledge for Interview Training
AI	Artificial Intelligence
API	Application Programming Interface
JWT	JSON Web Token
UI	User Interface
UX	User Experience
PDF	Portable Document Format
IDE	Integrated Development Environment
DBMS	Database Management System
SQL	Structured Query Language
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
SDK	Software Development Kit
UAT	User Acceptance Testing
RBAC	Role-Based Access Control
SSD	Solid State Drive
RAM	Random Access Memory
CPU	Central Processing Unit
ROI	Return on Investment

CHAPTER 1

INTRODUCTION

1.1 Background on GetPrep

With the current competitive and dynamic nature of the job market, landing a job involves more than mere academic credentials. Employers are now considering candidates more on the basis of communication skills, confidence, problem-solving skills, and cultural alignment—usually tested through the process of job interviews. Yet, numerous candidates experience fear and do not have the organized guidance to enable them to prepare well for such interviews.

This void has been filled by GetPrep interview platforms that are capable of replicating actual interview situations, offering instant feedback, and enabling users to continuously refine their performance. These platforms leverage developments in Artificial Intelligence (AI) to assess user responses— either verbal or textual—and offer personalized improvement suggestions.

The **GetPrep** Web Application is designed with the purpose of assisting students, job applicants, and professionals with interview practice in a virtual setting. Through dynamically generated role-specific interview questions and intelligent analysis of user answers, the system replicates the actions of an actual interviewer. It renders unnecessary costly training or manual mock interviewing by providing an automated, real-time, and scalable method.

By AI algorithms like Groq AI SDK Integration with Llama/Mixtral models for intelligent interactions. the system analyses key dimensions of a user's performance such as clarity, confidence, relevance, and technical accuracy. By integrating analytics and feedback systems, users are able to track their improvement and better prepare themselves for actual interviews under realworld conditions.

1.2 Scope of the Project

The **GetPrep** is conceptualized as an extensive digital solution enabling users to conduct actual interview sessions using the services of Artificial Intelligence. The scope of the project is to create a **user-friendly, scalable, and role-based web application that accommodates interactive interviews**, intelligent evaluations, and performance monitoring for users across various industries and experience levels.

Key Features within the Scope:

1. User Authentication & Security

Secure Access: Full registration and login flow using JWT (JSON Web Tokens) and crypt for password hashing.

Account Recovery: Integrated "Forgot Password" and "Reset Password" functionality.

2. Resume Management & Analysis

Advanced Upload: Interactive resume upload interface using react-drop zone.

Automated Parsing: Backend processing of PDF resumes to extract key information using PyPDF2 and AI-driven analysis.

3. AI-Powered Interview Simulation

Dynamic Interviews: Realistic interview sessions powered by the Grog AI (Llama/Mistral models).

Adaptive Questions: Questions that adapt based on the user's resume and previous answers.

Real-time Feedback: Instant evaluation of candidate responses.

4. Performance Analytics & Dashboard

Visual Insights: Interactive charts and performance metrics built with Recharts.

Tracking: A central dashboard to monitor progress and interview history.

5. Question Bank

A structured repository of common and role-specific interview questions for practice.

6. Reporting System

PDF Reports: Automated generation of detailed performance reports using Report Lab. Downloadable Insights: Users can download a summary of their interview performance for offline review.

1.2 Purpose and Objectives

The objective of this project is to create and develop an intelligent, interactive **GetPrep Web Application** that assists job applicants in practicing and improving their interview skills using AI-based simulations. This platform overcomes the drawbacks of conventional mock interviews—high costs, unavailability, and inconsistent feedback— by offering a scalable, customized, and automated solution based on **Artificial Intelligence (AI)** and **Natural Language Processing (NLP)**.

The project is designed to enable users with live practice, positive feedback, and tracking of their progress, hence boosting their confidence and preparedness for real job interviews. Through the simulation of realistic interview scenarios in different industries and levels of jobs, the application fosters inclusivity and flexibility for users with different professional aspirations.

Objectives of the Project:

1. Automated Interview Question Generation.

Provide improvement suggestions in communication, content quality, and overall performance.

2. User Progress Tracking and Analytics

- a. Integrate performance tracking features to track user development across several sessions.
- b. Utilize data visualization and scoring trends to enable users to recognize important areas for development.

3. Multi-Industry and Experience-Level Support

- a. Provide interview preparation support for different industries such as IT, finance, healthcare, engineering, and others.
- b. Create a platform that is accessible to students, new graduates, and experienced professionals.

1.3 Significance

The creation of an **GetPrep** Web Application is extremely important in the context of the current competitive and dynamic job market. Since job interviews are essential portals to career opportunities, preparation is more important than ever. Nevertheless, conventional interview coaching techniques tend to fail because of reasons such as restricted accessibility, prohibitive costs, and variable quality of feedback. This project deals with these challenges by taking advantage of the potential of Artificial Intelligence (AI) and Natural Language Processing (NLP) to offer a solid, affordable, and smart solution.

Areas of Importance

1. **Improved Interview Readiness:** The system provides a real-world and effective setting for job applicants to prepare through naturalistic, AI-simulated interviews. This makes it possible for users to become exposed to different patterns of questions and enhance their performance while responding under pressure.
2. **Increased Accessibility and Inclusivity:** In contrast to traditional mock interviews that necessitate human guides, this site operates 24/7 for anyone with access to the internet. This makes interview practice more accessible to students, graduates, and working professionals in urban and rural areas alike.
3. **Objective and Real-Time Feedback:** Through the application of NLP and AI based algorithms, the system is capable of objectively judging user responses. It offers immediate feedback on verbal delivery, tone, relevance of content, and confidence levels, allowing users to hone their skills step by step.
4. **Continuous Learning and Performance Monitoring:** The app monitors user improvement over multiple sessions, providing analytics of growth points and frequent mistakes. This ongoing learning strategy means users are not only practicing, but enhancing.
5. **Support of Multiple Industries and Experience Levels:** Whether a user is preparing for an entry-level IT job or a senior role in healthcare management, the system adapts accordingly. This multi-industry support ensures relevance and applicability across various career paths.
6. **Technological Innovation in Career Services:** The project illustrates how new technologies may be leveraged to reshape conventional education and career development paradigms. It demonstrates the application of AI not only for automating tasks, but for customized, smart learning experiences.

1.5. Overview of Methodology

The implementation of the **GetPrep Web Application** involves a systematic methodology to achieve systematic implementation, from design and planning to implementation and testing. The methodology integrates best practices in software engineering and AI-specific development approaches to design a scalable, interactive, and intelligent mock interview system.

Step-by-Step Overview of Methodology:

1. Requirement Analysis

- Collect and evaluate requirements from prospective users like students, career seekers, and professionals.
- Determine required features like dynamic question generation, instant feedback, and performance monitoring.

2. System Design

- Develop the system architecture with a proper separation between the frontend (user interface), backend (server logic), and AI/NLP modules.
- Implement navigation and responsiveness on various devices.
- Specify workflows like user registration, role assignment, interview session flow, and feedback submission.

3. Technology Stack Choice

- **Frontend:** HTML, CSS, React.js and Bootstrap for a responsive and dynamic interface.
- **Backend:** Node.js and Firebase to handle user sessions, data flow, and API interactions.
- **Database:** Firebase Fire store for handling interview data, user profiles, and session records.

AI-Driven Interview Simulation

- Allow users to choose job positions and industries to personalize the session.
- Dynamically create suitable questions using AI based on user choices.
- Permit text or voice responses, which are subsequently analysed with NLP for coherence, confidence, and technical correctness.

1. Real-Time Feedback and Scoring

- a. AI processes the response and gives immediate feedback on important parameters.
- b. Users are given customized recommendations for improvement in areas such as vocabulary usage, tone, technical content, and confidence.

2. Performance Tracking and Analytics

- a. Save each session's outcome and graph trends in the user's performance over time.
- b. Allow users to see strengths, determine repeating weaknesses, and redo sessions to enhance.

3. Testing and Validation

- a. Perform unit testing, integration testing, and user acceptance testing.
- b. Test the system's AI-provided feedback against expert-checked answers to verify quality and reliability.

4. Deployment and Maintenance

- a. Install the application on a cloud server with constant surveillance and upgrading.
- b. Maintain data privacy, user security, and scalability as user interest increase.

CHAPTER 2

FEASIBILITY STUDY

Prior to the development of any software system, it's crucial to evaluate if the project is feasible from different aspects. A feasibility study assists in establishing the viability of the project and ensuring that the suggested solution can be effectively designed, implemented, and maintained. This chapter analyses the **GetPrep** based on technical, economic, operational, legal, and time-based feasibility.

The feasibility study aids the development team to know the limitations, resources, and possible risks of the project. It makes sure that the project is compatible with technological capability, cost, and organizational objectives, as well as legal and ethical standards.

2.1 Technical Feasibility

Technical feasibility is concerned with whether the existing technology stack, development tools, and capabilities are sufficient enough to develop and maintain the GetPrep Interview Web Application.

Key Considerations:

- **Suitability of Technology Stack:** The technology stack used in the GetPrep Web Application is modern, scalable, and efficient for developing an AI-powered mock interview platform. React.js is used for building an interactive frontend interface, while Node.js, Express.js, REST APIs, and FastAPI handle backend processing and API integration. MongoDB Atlas and MySQL provide secure database management.

- **Resources Accessibility:** The software and platforms needed are opensource or commercially available. Cloud servers, contemporary web browsers, and development frameworks are all easily available.
- **Scalability and Performance:** With cloud hosting solutions, the application can be scaled according to user demand. The application of contemporary tools also guarantees high performance and minimal latency during real-time activities such as voice input processing through AI for instant feedback.
- **Ease of Integration:** The AI and NLP capabilities are API-based and modular, ensuring smooth integration with the core application. The APIs are capable of processing voice and text inputs, creating context based questions, and assessing user responses.
- **Team Expertise:** The technical skills necessary to develop this project is and it will (web development, API integration, database schema, and fundamental AI functions) are widely possessed by today's developers. Therefore, the technical team is well equipped to effectively execute the development process.

Conclusion:

The project is technically feasible. Through existing and available modern technologies, and with an able development team, the GetPrep app can be developed, deployed, and supported effectively.

2.2 Economical Feasibility

The economic feasibility looks at whether developing and maintaining the project is affordable based on the benefits that the project provides. In the **GetPrep Web Application** case, there is high economic feasibility. It is cheap to develop, particularly with contemporary open-source technology, and the gains far surpass what it takes in terms of investments.

Key Considerations:

1. Development Costs

- React.js is used for building an interactive and responsive frontend interface.
- Node.js and Express.js are used for backend development and server-side processing.
- AI and NLP-based technologies are integrated for intelligent interview question generation and response analysis.
- Real-time feedback and performance evaluation systems help users improve interview skills effectively.

2. Hosting and Maintenance

- Hosting is possible using cloud platforms Render with flexible pricing based on server usage.
- Maintenance is continuous server uptime, bug fixing, and periodic updates, which are low-cost and manageable.

3. Human Resources

- A small team with basic to intermediate web development and AI integration expertise can handle the entire system lifecycle.
- There is no requirement for costly or specialist AI engineers, so team expenses are low.

4. Scalability and User Reach

- One web-based application can support thousands of users without substantially raising costs, so it is very scalable.
- Once implemented, the platform can create long-term value with minimal monetary input.

5. Return on Investment (ROI)

- Saves customers money on expensive mock interview services or career consulting.
- Improves employment success rates, thus contributing indirectly to the financial security and employability of users.

Conclusion:

The GetPrep system is viable economically. With negligible infrastructure investment and maximum usability potential, the system promises great dividends to both the user and is developer in terms of a scalable and viable solution to preparing for interviews.

2.3 Operational Feasibility

Operational feasibility examines whether the system proposed addresses the problems identified to a great extent and to what extent it can be utilized effectively in a real-world setting. It considers whether the end users will embrace, utilize, and gain from the application when developed and implemented.

For the **GetPrep Web Application**, operational feasibility is high, because the platform is designed to address a distinct and prevalent problem: the need for lowcost, low-friction, and effective interview preparation.

Key Considerations:

1. User Acceptance and Accessibility

- The platform is made to be straightforward, easy-to-use, and available to a wide audience comprising students, freshers, and professionals.
- Web-based interface reduces installation requirements, and one can utilize it on any device and from any location.

2. Enhanced Learning Outcomes

- Real-time feedback and dynamic simulation interviews offer instant feedback on performance.
- Customized recommendations drive ongoing improvement, making the app a worthwhile tool for learning.

3. Use Case Adaptability

- Accommodates a wide range of industries (e.g., IT, finance, healthcare) and professional roles.
- Both beginners and advanced professionals can use it by adjusting the difficulty level and question context.

4. Minimal Human Supervision Required

- The GetPrep is able to perform interviews, analyse responses, and give feedback without any need for manual interference.
- This keeps the operation burden low and makes the system scalable.

Conclusion:

The system is feasible from an operational perspective. It represents an effective, scalable, and user-oriented solution that improves interview preparation by enabling intelligent automation. Its ease of use and suitability to address the needs of the user ensures successful adoption and long-term sustainability.

2.4 Legal Feasibility

Legal feasibility is the process of determining if the suggested project adheres to all the relevant laws, regulations, and moral principles. Legal feasibility for the **GetPrep Web Application** is critical so that user information is safe, intellectual property rights are upheld, and AI methods are open and responsible.

Key Considerations:

1. Data Privacy and Protection

- The system should adhere to data protection legislation like the General Data Protection Regulation (GDPR), particularly when dealing with user data from Europe.
- It should also adhere to India's Personal Data Protection Bill (once implemented) to maintain user privacy and safety of user data.

2. User Consent and Data Handling

- Explicit consent from users should be taken before storing responses or analysing performance.
- Adequate disclaimers should be given to ensure users that their data is not misused or shared.

3. Usage of AI and Open APIs

- AI services Groq API are utilized in the project, and these have licensing and usage terms of their own. Care must be taken to ensure that their usage complies with the providers' terms.
- No reverse engineering, misuse, or resale of AI services for commercial purposes should take place without the right authorization.

4. Ownership of Content and Intellectual Property

- AI-generated interview questions and user answers could be under intellectual property concerns.
- The system must establish ownership rights on user-provided content and AI-generated content.

5.Non-Discriminatory Algorithms

- AI systems should be designed and tested to prevent gender, ethnic, language accent, or other sensitive factor-biased judgments.
- Proper ethical AI practices must be adopted to promote fairness and transparency,

Conclusion:

The GetPrep Web Application is legally viable, as long as it includes proper privacy policies, ethical standards of AI usage, and adheres to local and global data protection legislation. By addressing these areas pre-emptively, the project can sidestep legal issues and gain user trust.

2.5 Timeline Feasibility

Timeline feasibility assesses if the project is feasible to be delivered within a reasonable and defined timeframe, given the scope, resources, and complexity. For the **GetPrep Web Application**, the suggested timeline is realistic and well-structured, with opportunities for consistent progress throughout each phase of development.

The project duration is estimated to be 12 weeks, which is adequate for analysis, design, development, testing, and deployment.

Suggested Timeline Breakdown:

PHASE	TIME DURATION
Requirement Analysis	1 Weeks
Design Phase	2 Weeks
Development Phase	6 Weeks
Testing phase	2 Weeks
Phase	1 Weeks
Total Duration	12 Weeks

Supporting Factors:

- **Clear Task Division:** Each phase clearly has deliverables and can be performed in parallel by separate team members where feasible.
- **Moderate Complexity:** The utilization of APIs and pre-existing frameworks eliminates excessive workload and minimizes development.

Conclusion: The GetPrep Web Application is within the suggested timeline and feasible. With effective planning and task delegation, the system can be developed and rolled out in full within 18 weeks, thus appropriate for academic, prototype, or startup- level implementation.

CHAPTER 3

PROJECT OBJECTIVE

For any software project, properly defining the objectives is an important step in guaranteeing successful development and deployment. The objectives give direction, lead the design and implementation stages, and assist in measuring success of the system upon completion. For the **GetPrep Web Application**, the overarching goal is to provide an intelligent, scalable, and interactive system that helps individuals prepare for job interviews by leveraging Artificial Intelligence (AI) and Natural Language Processing (NLP)-powered realistic simulations.

The current chapter elucidates the large-scale vision of the project and discusses the important and specific goals driving its implementation.

3.1 Key Objectives

The **principal goals** of this project revolve around addressing actual life problems of job candidates in getting ready for interviews. Conventional mock interviews are in need of human mentors, and this may prove to be expensive, time-consuming, and challenging to scale. This project focuses on personalizing and automating the interview preparation with the aid of AI technology.

Core Objectives:

1. Emulate Real Interview Scenarios:

- Present users with profession-based questions and real-like interview flow.
- Imitate both technical and HR interviews in different industries.

2. Apply AI for Adaptive Interviewing:

- Dynamically create questions depending on user profile, job, and experience.
- Adapt complexity in real time according to the responses of the candidate.

3. Provide Instant, Personalized Feedback:

- Employ AI to evaluate verbal and written answers.
- Offer advice on clarity, relevance, confidence, and technical accuracy.

4. Monitor Progress and Performance:

- Provide metrics that demonstrate improvement over time.
- Direct users to improve their weak points with specific feedback

5. Make It Scalable and Accessible:

- Create a web-based solution accessible to anyone who has an internet connection.
- Accommodate a large base of users from students and recent graduates to office workers.

3.2 Project Objectives

The project goals set the key objectives that dictate the development, functionality, and purpose of the **GetPrep Web Application**. These goals are set to ensure the system provides an authentic, smart, and highly effective platform for interview practice. Every goal is intended to address certain issues that job seekers experience, including a lack of access to customized mock interviews, minimal feedback, and the requirement of scalable solutions.

1. To Create an Intelligent Mock Interview System

The main objective is to create an intelligent web application capable of mimicking real job interviews. The system should dynamically formulate interview questions depending on the chosen job category and experience level of the user. This system, unlike traditional question banks, has to change in real time, providing a different experience for every user. The simulation should be both technical and behavioural in style, replicating what candidates experience in real interviews.

2. For Integrating NLP and Artificial Intelligence

The app will make use of Artificial Intelligence and NLP to monitor the verbal and text responses from the user. The AI tool must be able to assess measures like clarity, coherence, grammar, confidence, voice tone, and technical correctness.

This combination is key to the platform being able to provide automated, unprejudiced, and knowledgeable feedback—something lacking in customary mock interviews.

3. To Give Real-Time Feedback and Performance Assessment

One of the most distinctive features of this project is that it can provide real-time automated feedback. The system should rate the user after every question according to pre-set criteria and offer suggestions for improvement. This will make users adapt and rephrase their answers immediately, which will speed up their learning and make them more ready for real interviews.

4. To Create an Intuitive, Responsive User Interface

The platform will provide a smooth user experience. With React.js and contemporary UI/UX principles, the interface will be made clean, responsive, and intuitive. It should accommodate access from multiple devices—desktops, tablets, and smartphones—so that users can practice interviews anywhere, anytime.

5. To Accommodate Multi-Industry and Experience-Level Customization

The application needs to appeal to a broad set of users through customization by industry (e.g., IT, Finance, Healthcare, Engineering) and user type (e.g., student, fresher, experienced professional). This makes the platform useful and relevant to a broad set of users with diverse requirements.

6. To Track User Progress and Learning Trends

The system will need to remember session data and offer performance metrics to users. This entails tracking scores against previous scores, uncovering frequent errors, and underlining areas which require extra study. Users can be made to go back through past sessions to compare progress by encouraging continuous learning.

7. In Order to Have Security, Privacy, and Compliance

User information—such as responses, feedback, and personal details—needs to be stored and processed securely. The platform needs to comply with applicable data privacy legislation and best practices, providing users with transparency regarding how their data is processed.

CHAPTER 4

HARDWARE AND SOFTWARE REQUIREMENTS

4.1. Hardware Requirements

Hardware specification is essential in order to assure smooth performance for development, testing, and deployment.

These details are classified below:

- **Development and Testing Environment**

Processor: A Quad-Core processor (Intel i5 or AMD Ryzen 5) to be used for effective multitasking while developing.

RAM: 8 GB of RAM as the minimum, but 16 GB for enhanced speed, particularly when multiple tools (IDE, browsers, and testing environments) are being used concurrently.

Storage: 50 GB of available disk space, ideally on an SSD, for rapid data access and uninterrupted use.

Network: A high-speed internet connection (at least 10 Mbps) is needed for real-time voice interactions and cloud access. **Audio Devices:** Good speakers and microphone are required for voice-based interview interactions.

Production Server Requirements

Processor: Multi-core processors (Intel Xeon or AMD EPYC) are advised for serving multiple users.

1. **RAM:** 16 GB, scalable according to user needs.
2. **Storage:** 100 GB SSD (expandable) for data storage and fast access.
3. **Network:** High-speed internet
4. **Security:** SSL Certificates, Firewall Protection.

4.2 Software Requirements

The GetPrep Platform's software stack is built for scalability, security, and effective development. It consists of:

- **Operating System:** Supports Windows 10/11, macOS, and Linux (Ubuntu 20.04+). This cross-platform compatibility provides ease for developers.
- **Development Tools**
 - IDE:** Visual Studio Code or WebStorm for code management with syntax highlighting along with inbuilt Git support.
 - Version Control:** Git for source code management, with GitHub as the collaborative development platform.
 - Deployment Platform:** Vercel for front-end deployment (Next.js) and Firebase Console for backend services.

- **Programming Languages**

React.js & Node.js: React.js for front-end development, while Node.js for Back-end.

- **Frameworks and Libraries**

1. Frontend:

- HTML5 is used to structure the web pages and define the layout of the application. It provides semantic elements that improve readability, maintainability, and search engine optimization..
- CSS3 is utilized for designing and styling the user interface. It enhances the visual appearance of the application through responsive layouts, animations, transitions, and flexible design components.
- React.js is the primary frontend framework used in the project. It enables the development of reusable UI components and improves application performance through virtual DOM rendering.

2. Backend:

The backend of the GetPrep Web Application is designed to manage business logic, authentication, database communication, and API handling efficiently. It ensures secure data processing, scalability, and smooth interaction between the frontend and database systems.

•Node.js

Node.js is used as the runtime environment for executing server-side JavaScript code. It provides fast performance, non-blocking architecture, and efficient handling of multiple user requests simultaneously.

•Express.js

Express.js is used as the backend framework to simplify server-side development. It helps in creating RESTful APIs, routing requests, handling middleware, and managing server operations effectively.

•REST API

REST APIs are implemented to establish communication between the frontend and backend. These APIs enable secure data exchange, user authentication, interview management, and retrieval of personalized recommendations.

•Fast API

Fast API is used for implementing high-performance APIs and AI-related functionalities within the application.

Database Technologies Used

The GetPrep Web Application uses modern database technologies to efficiently store, manage and retrieve user and application data. The database system ensures data security, scalability, reliability and fast access to information.

•MongoDB

MongoDB is a NoSQL database used to store unstructured and semi-structured data such as user activities, interview records, recommendations, and application logs. It provides high scalability, flexibility and fast data retrieval.

•MySQL

MySQL is a relational database management system used to store structured data such as user credentials, profiles, course details, and administrative records. It ensures data consistency, integrity, and secure transaction management.

•DatabaseConnectivity

Database connectivity is established using secure APIs and backend services to ensure smooth communication between the application and database servers.

•DataSecurity

The database system includes authentication, authorization, encryption, and backup mechanisms to protect sensitive user information and maintain data reliability.

Features of Database System

- Efficient storage and retrieval of data
- Secure management of user information
- High scalability and flexibility
- Fast query execution
- Reliable backup and recovery support
- Structured and unstructured data handling

CHAPTER 5

PROJECT FLOW

5.1 Problem Identification

With the current digital-first job market, the necessity for organized, role-based interview preparation has never been more important. Yet, conventional approaches like face-to face mock interviews, coaching classes, and peer sessions have a number of drawbacks:

- **Accessibility Issues:** Not all individuals can afford experienced mentors or professional coaching services, especially students in remote or underserved communities.
- **Steep Price:** Custom coaching and HR advice is costly, not affordable for many job seekers.
- **Unchanging Learning Architectures:** All but a handful of existing web-based mock interviewing systems provide inflexible question lists that don't change based on user performance, profession, or level of experience.
- **Insufficient Live Feedback:** There is often little or no instantaneous, actionable feedback. Users never know what went wrong or how to correct their mistakes.
- **Limited User Roles:** Solutions tend to equal all users when they are not a fresher, an experience professional, or a career-switcher.

This project aims at creating a solution that addresses all these limitations with the introduction of an **GetPrep** web application that is inclusive, scalable, smart, and personalized to every user's need.

5.2 Requirement Analysis

An in-depth requirement analysis was done to make sure that the system is catering to users' needs as well as being technically feasible and easy to use. The requirements are categorized under functional and non-functional.

Functional Requirements:

The system is supposed to accommodate the following main functions:

- **User Registration & Login:** Users must be able to register, log in, and manage their profiles safely.
- **Role Selection:** When users register, they should choose their role—Job Seeker, Student, or Professional—to allow the interview flow to be customized accordingly.
- **Industry & Job Type Selection:** Users should be able to choose their industry (e.g., IT, Finance, Healthcare) and job type (e.g., Developer, Analyst, Manager).
- **Dynamic Question Generation:** The system should employ AI in generating questions from chosen job roles and experience levels.
- **Input Modes:** Users must have the option of responding to interview questions through voice or text.
- **AI Evaluation:** User responses must be evaluated by analysing them using NLP methods as a measure for clarity, tone, confidence level, and correctness.
- **Progress Tracking:** Users must see analytics reflecting the trends in performance, repeated errors, and skills to improve upon.

Non-Functional Requirements:

- **Scalability:** The system should support 100+ simultaneous users without slowing down.
- **Responsiveness:** The interface needs to adjust to various device types and screen sizes (mobile, tablet, desktop).
- **Security:** Secure login protocols like JWT (JSON Web Tokens) and encryption of data must be used.

- **Performance:** The system needs to react to user actions and provide feedback in 1–2 seconds for a delightful experience.
- **Accuracy:** Evaluations done by AI should have an accuracy of no less than 85% with respect to human evaluators.

5.3 Design Phase

The design phase concentrates on organizing the whole application flow—both technically and visually—to make sure that users enjoy a seamless, intuitive experience while preserving system integrity and performance.

A. User Interface Design:

- Built using React.js to enable a dynamic and responsive UI.
- Interfaces differ according to user roles to customize the experience.
- Important screens are:
- Login/Signup Page
- Role & Industry Selection
- Interview Dashboard
- Response Input Page
- Feedback & Analytics Page
- Utilization of icons, color-coded feedback, and charts to present insights in a visually appealing manner.

B. System Architecture:

- The architecture is of a client-server model, segregated into:
- Frontend (React.js): Manages user interface and interaction.
- Backend (Node.js): Handles server-side logic, routing, and API interactions.
- Database (MySQL & MongoDB): Saves user information, responses, scores, and feedback history.

C. Database Modelling:

Important entities: Users, Interview Sessions, Questions, Responses, and Feedback.

Relations:

- A User may have multiple Interview Sessions.
- Every Interview Session is associated with several Questions and Responses.
- Data is normalized to provide quick querying and low redundancy.

D. Security Design:

- Passwords are hashed using hashing algorithms.
- Role-based access control (RBAC) provides users with access to only relevant data.
- Input validation avoids injection attacks and maintains data integrity.

5.4 Technology Selection

Selecting the appropriate technologies was instrumental to the success of the GetPrep IT Web Application. The system needed to be reliable, responsive, scalable, and able to facilitate AI-processing in real-time. On these bases, the technology stack was chosen.

A. Frontend Technologies

- **Html5**

Used for creating and structuring web pages of the GetPrep Web Application.

- **CSS3**

Used for styling, responsive layouts, animations, and enhancing the visual appearance of the application.

- **JavaScript**

Used for adding dynamic functionality, form validation, user interaction handling and API integration.

- **React.js**

Used for building reusable UI components and developing a fast, interactive single-page application.

- **Bootstrap**

Used for responsive design, prebuilt UI components, and improving frontend development efficiency.

- **AxiosAPIIntegration**

Used for handling API requests and enabling communication between frontend and backend services.

Features of Frontend System

- Responsive and user-friendly interface
- Interactive UI components
- Fast page rendering
- Cross-device compatibility
- Real-time user interaction handling
- Smooth navigation and performance

B. Backend Technologies

- **Node.js**

Used as the server-side runtime environment for executing backend JavaScript code efficiently.

- **Express.js**

Used for creating RESTful APIs, handling routes, middleware, and server-side operations.

- **RESTAPI**

Used for secure communication and data exchange between frontend and backend systems.

- **FastAPI**

Used for high-performance API development and AI-based functionalities in the application.

- **JWTAuthentication**

Used for secure user authentication, authorization, and session management.

Features of Backend System

- Secure authentication and authorization
- Fast API response handling
- Scalable server-side architecture
- Efficient database connectivity
- Real-time data processing
- High performance and reliability

C. Database Management

- **MongoDBAtlas**

Used as a cloud-based NoSQL database for storing unstructured and semi-structured data efficiently.

- **MySQL**

Used as a relational database management system for handling structured data and maintaining data consistency.

- **FirebaseAuthentication**

Used for secure user authentication, login management, and session handling.

- **DatabaseConnectivity**

Used to establish secure communication between backend services and database systems.

- **CloudDatabaseServices**

Used for scalable storage, real-time synchronization, and reliable database performance.

- **DataSecurityMechanisms**

Used for encryption, backup, authentication, and protection of sensitive user information.

Features of Database System

- Secure data storage and management
- Fast data retrieval and query execution
- High scalability and cloud support
- Backup and recovery support
- Real-time database synchronization
- Efficient handling of structured and unstructured data.

5.5 Development

Development of the **GetPrep** Web App was carried out in iterative cycles to allow for ongoing testing and feedback. The primary steps of the development life cycle are as follows:

A. Phase 1 – Environment Setup

- Install local development environment with:
- **IDE:** Visual Studio Code or WebStorm for code management with syntax highlighting along with inbuilt Git support.
- **React.js** (React-based) for dynamic user interface, Tailwind CSS for responsive design
- Firebase for real-time database management and secure authentication.

B. Phase 2 – User Authentication & Role Management

- Developed registration and login pages with role-based redirection.
- Implemented JWT-based secure authentication.

- Established three primary roles:
 - Job Seeker
 - Student
 - Professional

Each role was provided with customized dashboard access and interview options.

C. Phase 3 – Interview Session Integration

- Developed interfaces for:
 - Choosing industry and job role.
 - Starting mock interview session.
- Developed API endpoints to:
 - Call the AI model and dynamically create questions.
 - Receive and cache user answers (text or voice).
 - Retrieve AI feedback and show scores.

D. Phase 4 – AI Evaluation Integration

- Integrated Groq APIs to evaluate answers and provide:
 - Clarity, confidence, and relevance scores.
 - Improvement suggestions.
- Employed asynchronous API calls to keep the app responsive when waiting for results generated by AI.

E. Phase 5 – Feedback & Analytics

- Created a dashboard

to display:

- Session scores for individual sessions.
- Graphical charts of performance trends.
- Most common errors.
- Made it possible to filter performance history by:
 - Date
 - Type of job
 - Score type (clarity, content, etc.)

F. Phase 6 – Responsive Design & UX Improvements

- Made mobile compatibility a guarantee.
- Included loading indicators, toast notifications, and error management.
- Incorporated Framer Motion animations for better transitions and interactions.

G. Phase 7 – Testing and Refinement

- Employed unit tests to guarantee frontend and backend module reliability.
- Manually tested every user flow.
- Optimized API performance and reduced response latency.

5.6 Testing and Validation

Testing is an important step in the software development life cycle. It verifies that the system works correctly, securely, and reliably in various scenarios and user roles. The GetPrep Web Application went through various testing approaches, using manual and automated testing to confirm its behaviour from both functional and non-functional aspects.

A. Unit Testing

- Centred on standalone modules like:
 - User registration and login.
 - API endpoints for interview generation.
 - AI feedback and scoring evaluation.
- Tools utilized:
 - Jest for testing JavaScript.
 - Postman for testing backend APIs.

B. Integration Testing

- Verified the integration between:
 - Frontend elements and backend APIs.
 - AI response processing and user sessions.
- Verified that:
 - AI-generated questions were displayed correctly.
 - User-submitted responses were correctly analysed and saved.

C. Functional Testing

Confirmed the system's functionality against the specified requirements:

- Dynamic question generation.
- Role-based navigation.
- AI feedback integration.
- Analytics dashboard display.

All user flows (Student, Job Seeker, Professional) were separately tested to ensure that role-based features functioned as expected.

D. User Acceptance Testing (UAT)

Performed using real users (students and colleagues).

- Got good feedback on:
 - Ease of interface.
 - Realism in quality of AI-generated questions.
 - Utility of the feedback provided.
- Improved the following based on user feedback:
 - Improve handling of voice inputs.
 - Offering more descriptive improvement hints.

E. Validation of AI Feedback

- Compared AI feedback outcomes with expert ratings.
- Made sure that:
 - Scoring of "confidence" and "clarity" aligned with 85% of human evaluations.
 - Correctly identified content relevance, particularly for technical positions.
- Adjusted AI prompts and parameters to optimize model output.

F. Performance Testing

Evaluated system behaviour under load:

- Simulated multiple users creating sessions simultaneously.
- Guaranteed backend API response time at <2s.
- Optimized database queries and front-end rendering to minimize lag.

G. Bug Tracking and Resolution

- Issues were tracked with Trello and GitHub Projects.
- Common bugs included:
 - Delay in rendering feedback.
 - AI returning overly generic advice.

5.7 Authentication Flow:

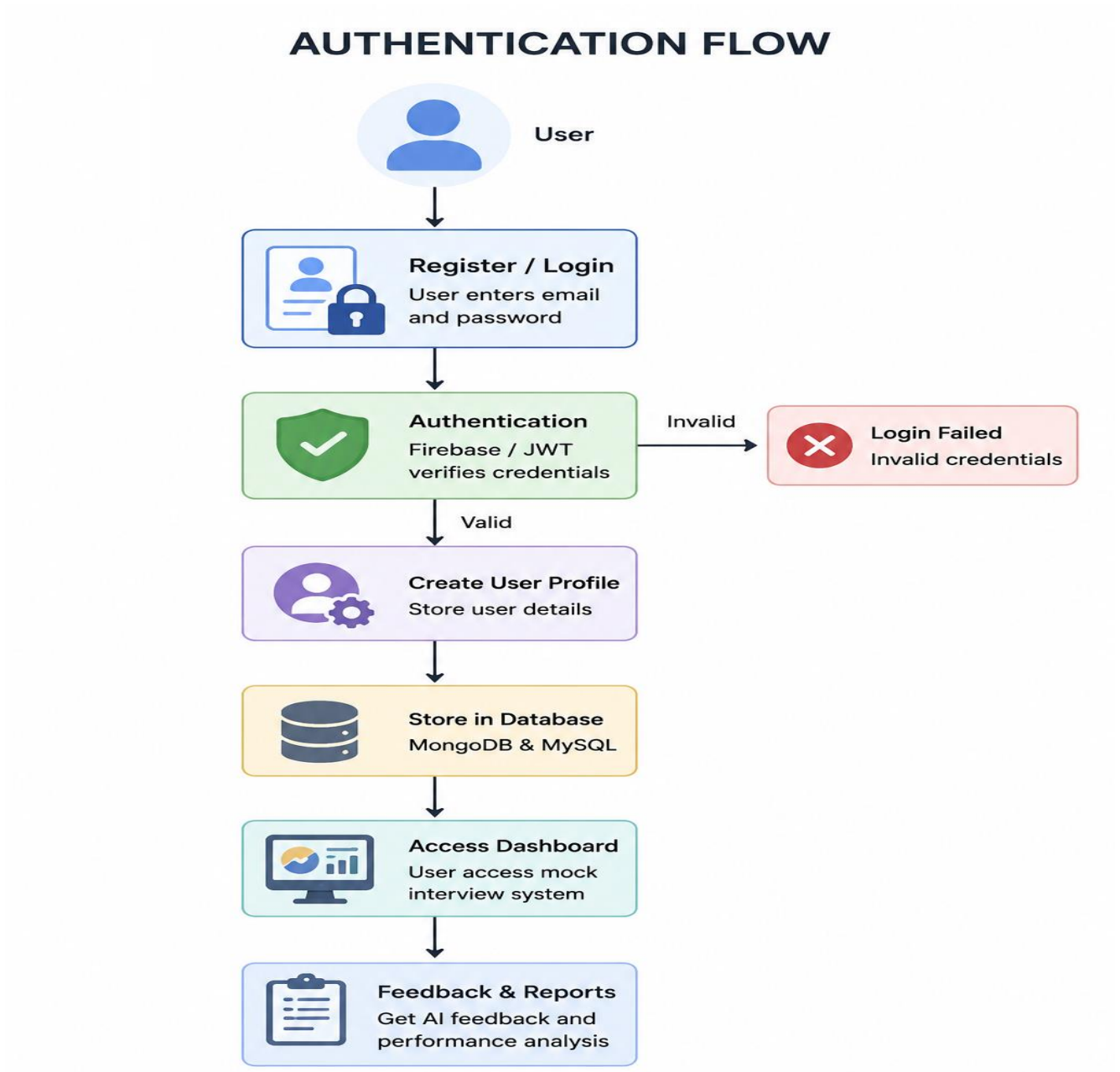


Fig. 5.7.1 Authentication Flow

Explanation of the Flow:

1. User Registration

The user begins the registration process by entering personal details such as name, email address, password, and other required information to create an account in the GetPrep Web Application.

2. User Login Authentication

After registration, the user logs into the system using valid email credentials and password. The authentication system verifies the user identity to ensure secure access to the platform.

3. Backend Authentication Process

- The backend authentication system using Node.js, Express.js, JWT Authentication, and Firebase Authentication validates user credentials securely.
- If the entered credentials are correct, the user is authenticated successfully and granted access to the application dashboard.

4. Create User Profile

- After successful authentication, a personalized user profile is created automatically.
- The profile stores important user information such as name, interview preferences, progress records, skills, and performance analytics.

5. Store User Data in Database

- User information and profile details are securely stored in MongoDB Atlas and MySQL databases.
- MongoDB stores dynamic and unstructured data such as interview history and recommendations, while MySQL stores structured user records and authentication details.

6. Access Mock Interview System

- Authenticated users can access mock interviews, aptitude tests, technical questions, and AI-based interview preparation features.
- The system provides personalized interview experiences and tracks user performance in real time.

7. Real-Time Feedback and Recommendations

The application analyzes user performance and provides instant feedback, improvement suggestions, and personalized recommendations for better interview preparation.

Relation to GetPrep Project

- This authentication and database flow enables secure registration, login, and personalized access to the GetPrep mock interview platform.
- The system securely stores user data and allows users to continue their interview preparation journey seamlessly.
- The integration of frontend, backend, APIs, and cloud databases ensures scalability, security, and smooth application performance.
- The authentication system protects user information and ensures that only authorized users can access interview preparation features and personal records.

5.8 Interview Workflow:

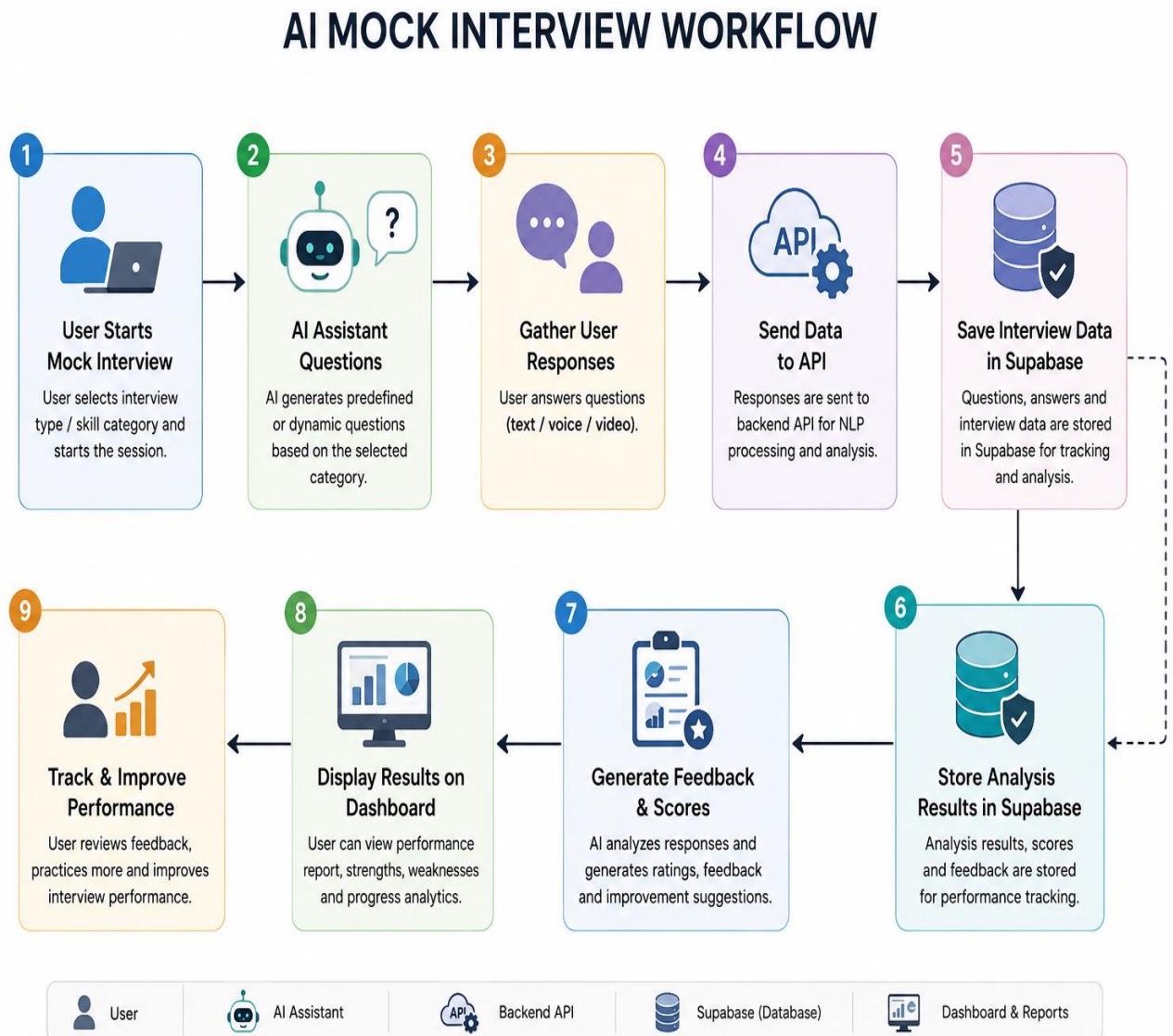


Fig. 5.8.1 Interview Workflow

Workflow Explanation:

The GetPrep Web Application is an AI-powered interview preparation platform designed to help students, freshers, and professionals improve their interview skills through real-time mock interview sessions.

The workflow of the system explains how users interact with the platform, how interview Sessions are conducted, and how feedback and performance analysis are generated.

The system integrates frontend technologies, backend APIs, authentication services, AI processing, and cloud databases to ensure smooth and secure operation.

The interview workflow is designed to simulate a real interview environment where users Can Practice technical, HR, aptitude, and behavioral interviews. The application provides automated question generation, response analysis, instant feedback, and personalized recommendations to improve interview performance.

Detailed Workflow Explanation**1. User Registration and Login**

The workflow begins when a user accesses the GetPrep Web Application and creates an account using registration functionality. The user provides details such as name, email address, password and other required information.

After registration, the user logs into the system using secure authentication mechanisms implemented using Firebase Authentication and JWT Authentication. The authentication system verifies user credentials and allows only authorized users to access the interview platform.

Main Functions:

- User account creation
- Secure login authentication
- User session management
- Protected dashboard access

2. User Dashboard Access

After successful login, the user is redirected to the dashboard of the application. The dashboard acts as the central interface from where users can access different interview preparation modules.

The dashboard displays:

- Available interview categories
- Previous interview history
- Performance reports
- Personalized recommendations

3. Interview Category Selection

The user selects the preferred interview type based on career goals and preparation requirements.

Different interview modules available in the system include:

- Technical Interviews
- HR Interviews
- Aptitude Tests
- Behavioral Interviews
- Coding Interviews
- Communication Skill Assessments

The selected category helps the AI system generate relevant interview questions dynamically.

4. AI-Based Question Generation

Once the interview category is selected, the AI-powered interview assistant starts generating questions. Questions may be predefined or dynamically generated using AI and Natural Language Processing (NLP) techniques.

The system analyzes:

- User-selected category
- Skill level
- Experience level

- Previous interview performance

Based on this analysis, suitable interview questions are displayed one by one to simulate a real interview environment.

Features:

- Dynamic question generation
- Real-time interaction
- Personalized interview experience
- Domain-specific question handling

5. User Response Submission

The user answers interview questions through text input or voice input depending on application implementation. The responses are collected securely and temporarily processed before analysis.

The system supports:

- Text-based responses
- Voice-based responses
- Real-time answer submission
- Continuous interview interaction

This stage plays an important role in evaluating the communication and technical abilities of the user.

6. Backend API Processing

After collecting user responses, the application sends the data to backend APIs developed using Node.js, Express.js, and FastAPI.

The backend APIs perform several important operations:

- Validate interview data
- Process user responses
- Manage interview sessions
- Communicate with databases
- Trigger AI analysis modules

FastAPI is used for handling AI-related functionalities because of its high

performance and asynchronous processing capabilities.

7. AI and NLP-Based Analysis

The AI analysis module processes user responses using machine learning and NLP techniques.

The system evaluates:

- Technical correctness
- Communication skills

The AI engine generates intelligent feedback based on the user's performance.

Advantages:

- Automated evaluation
- Instant performance analysis
- Personalized recommendations
- Continuous improvement support

8. Database Storage and Management

All interview-related information is securely stored in MongoDB Atlas and MySQL databases.

MongoDB Atlas Stores:

- Interview records
- User activities
- Dynamic reports
- AI-generated feedback
- Interview history

MySQL Stores:

- User credentials
- Authentication details
- Structured profile information
- Administrative data

The database architecture ensures:

- High scalability

- Secure data storage

9. Real-Time Feedback Generation

After AI analysis is completed, the system generates instant feedback reports for the user. The feedback includes:

- Overall performance score
- Technical skill analysis
- Communication evaluation
- Areas for improvement
- Personalized learning suggestions

The feedback system helps users identify weaknesses and improve interview preparation effectively.

10. Performance Dashboard and Analytics

Finally, all interview reports and analytics are displayed on the user dashboard.

Users can Monitor. Their progress over time and analyze previous interview sessions.

The dashboard provides:

- Graphical performance reports
- Interview history tracking
- Skill improvement analytics
- Personalized preparation roadmap

This helps users continuously improve their interview performance and build confidence for real-world job interviews.

Conclusion

The interview workflow of the GetPrep Web Application demonstrates the complete functioning of an AI-powered interview preparation system. The workflow integrates frontend technologies, backend services, AI processing, authentication systems, APIs, and cloud databases to create a secure, scalable, and intelligent interview platform. The system provides a real-time mock interview environment, automated response evaluation, and personalized feedback mechanisms that help users improve technical knowledge, communication skills, and overall interview confidence.

CHAPTER 6

PROJECT OUTCOME

6.1 Login Pages

Description:

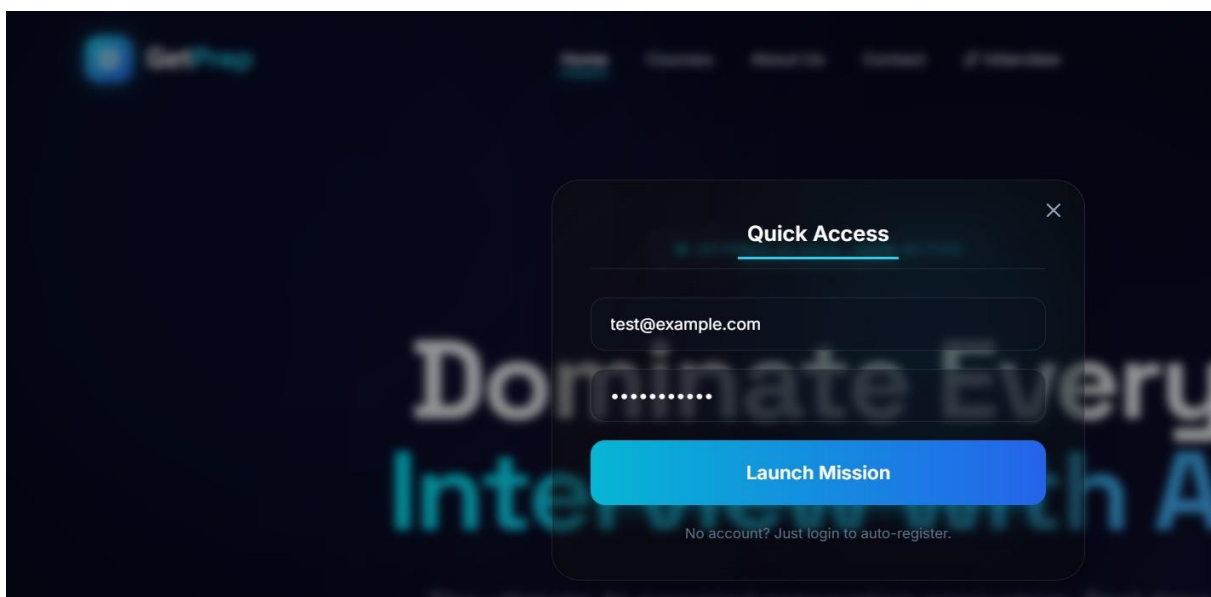
This screen represents the authentication module of the system where users can securely log in using their email and password.

Key Features:

- User-friendly login interface
- Email and password input fields
- Password visibility toggle option
- “Forgot Password” functionality
- Redirect option for new users to register

Purpose:

To authenticate users and provide secure access to the platform.



Description:

This dashboard is shown when a new user logs in and has not yet attempted any mock interviews.

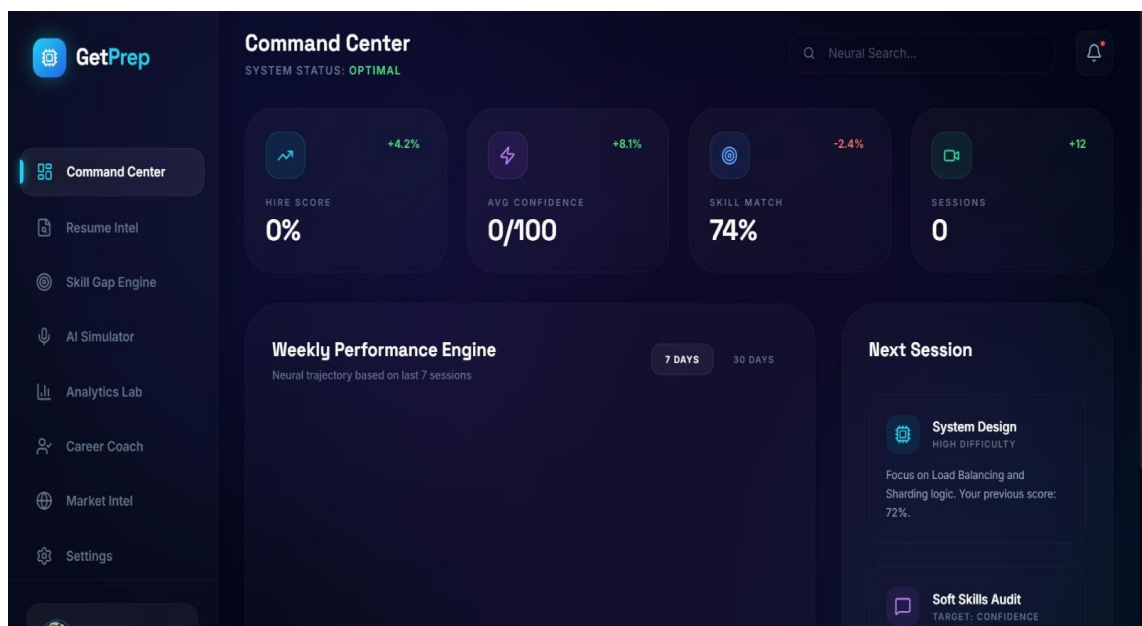
Key Features:

- Displays zero activity metrics
- Sections for performance trends and competency breakdown
- Call-to-action button: “Kick off your first interview”

Purpose:

To guide new users and encourage them to start their first mock interview.

- **Dashboard (With Data)**



This screen shows the user’s performance after attempting interviews.

Key Features:

- Total interviews, average score, best score, weekly sessions
- Score performance trend graph
- Competency breakdown chart
- Recent activity table with interview details

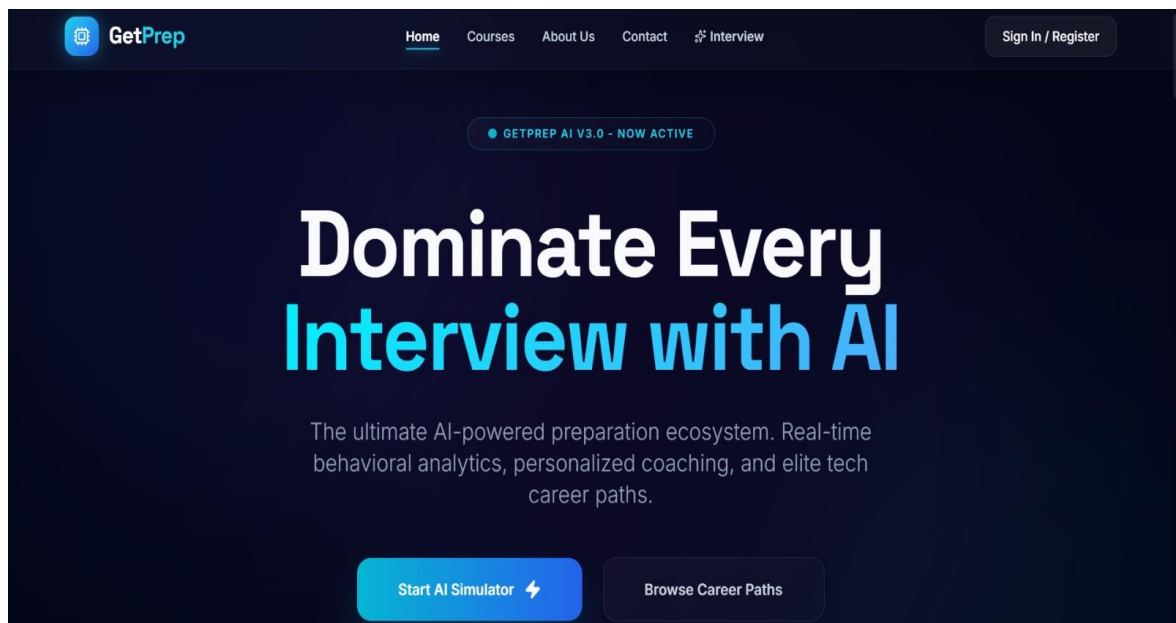
Purpose:

To provide insights into user performance and progress tracking.

- **Mock Interview Setup Page**

Description:

This module allows users to configure and start a mock interview session.

**Key Features:**

- Interview types: HR, Technical, Behavioral, System Design, Aptitude, Resume-based
- Difficulty levels: Easy, Medium, Hard
- Filters for company and profile
- “Start Mock Interview” button

Purpose:

To personalize interview preparation based on user preferences.

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